ENGINEERING PORTFOLIO

Nathaniel Murcin

Structural and fluid systems design, from concept to hardware.

Georgia Institute of Technology
B.S. Aerospace Engineering, expected May 2027
M.S. Aerospace Engineering, expected May 2028

BLUE ORIGIN · GTXR · ZINN COMBUSTION LAB · PURSUIT

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Cryogenic tank-farm P&ID design

Blue Origin · Fluids & Manufacturing Engineering Intern · May–Aug 2026 · Cape Canaveral, FL

PROBLEM

- Design a critical-path cryogenic tank-farm P&ID
- Starting point: heritage designs

CRITERIA

- Meet the facility's requirements with the fewest components
- Every line, valve and relief device sized and validated
- Hazards analyzed before formal HAZOP review

DECISION

- Challenged requirements, eliminated branches and downsized hardware
- Sized, selected and validated cryogenic piping, valves and relief devices: Cv, Darcy-Weisbach, line sizing, overpressure relief
- Developed the BOM and coordinated suppliers

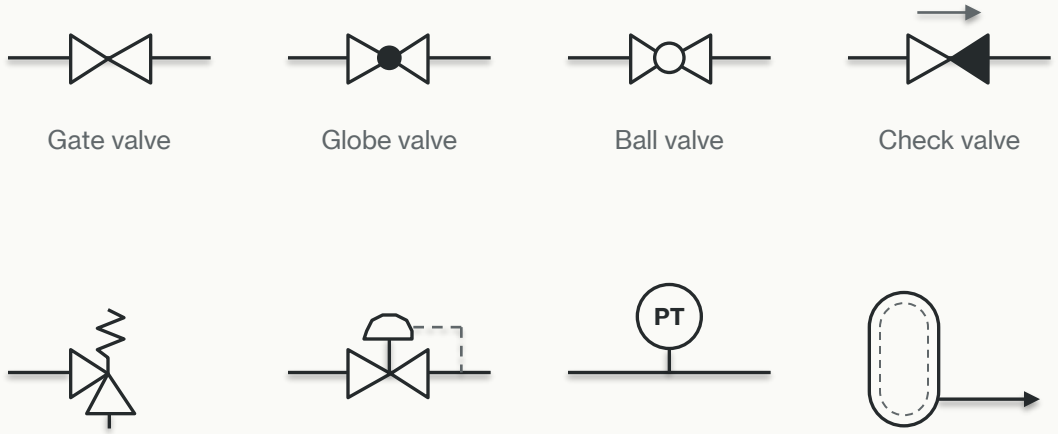
VERIFICATION

- Preliminary hazard analysis: component failure modes and consequences against safeguards
- Documented the CONOPS for formal HAZOP review

~60%

fewer components than heritage designs

P&ID SYMBOLS (GENERIC)



Generic P&ID symbols for illustration, not the Blue Origin system. No Blue Origin hardware, facilities, drawings or specifications are shown.

Lunar tooling structural analysis

Blue Origin · Fluids & Manufacturing Engineering Intern · May–Aug 2026 · Cape Canaveral, FL

PROBLEM

- Grip lunar hardware without part slippage or damage

CRITERIA

- Positive margins against yield and ultimate for every load case

DECISION

- Analyzed the fixture **like a flight structure**
- Hand calculations for each load case, cross-checked with Creo FEA

VERIFICATION

- Each load case evaluated against the yield and ultimate margin criteria

STRUCTURAL ANALYSES PERFORMED

Torque resistance	Bolt shear & tension
Preload	Joint slip
Tear-out	Bending
Thermal expansion (CTE)	Creo FEA cross-check

No Blue Origin hardware images or specifications are shown.

Designing the fins

Fin Design & Manufacturing Lead, Georgia Tech Experimental Rocketry · Aug 2023–present

PROBLEM

- Early fin architecture needed two decisions: cross-section geometry and core material

CRITERIA

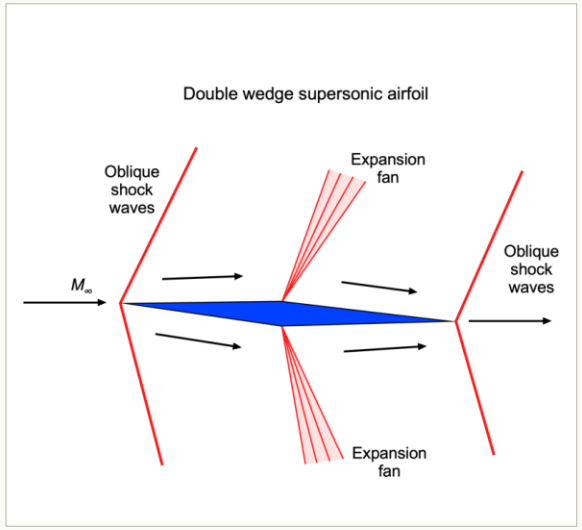
- Manufacturability & repeatability
- Mass
- Aerodynamic performance
- Process safety

DECISION

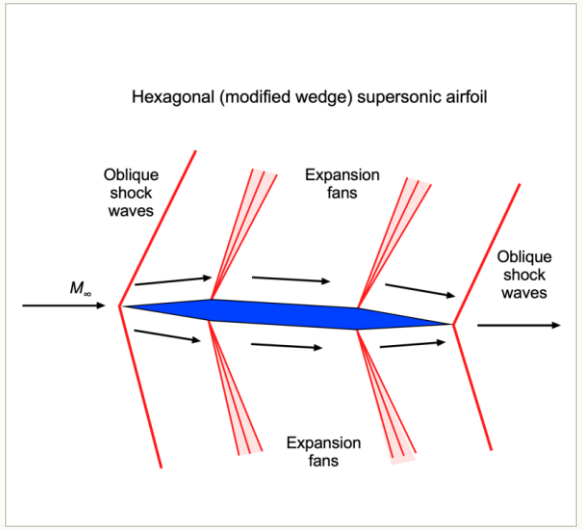
- Goal: an architecture that can be **built consistently across eight flight fins**, rather than one that maximizes single-fin performance

VERIFICATION

- Simulation results informed the architecture decisions
- Ownership boundary: detailed geometry optimization was done later by the GTXR simulations team



Double wedge
Aerodynamically preferred; harder to machine repeatably



Hexagonal
Repeatable to manufacture; higher predicted flutter margin

Shock diagrams: Introduction to Aerospace Flight Vehicles, Embry-Riddle Aeronautical University.

Decision tree trade study

PROBLEM

- Choose the fin cross-section and taper

CRITERIA

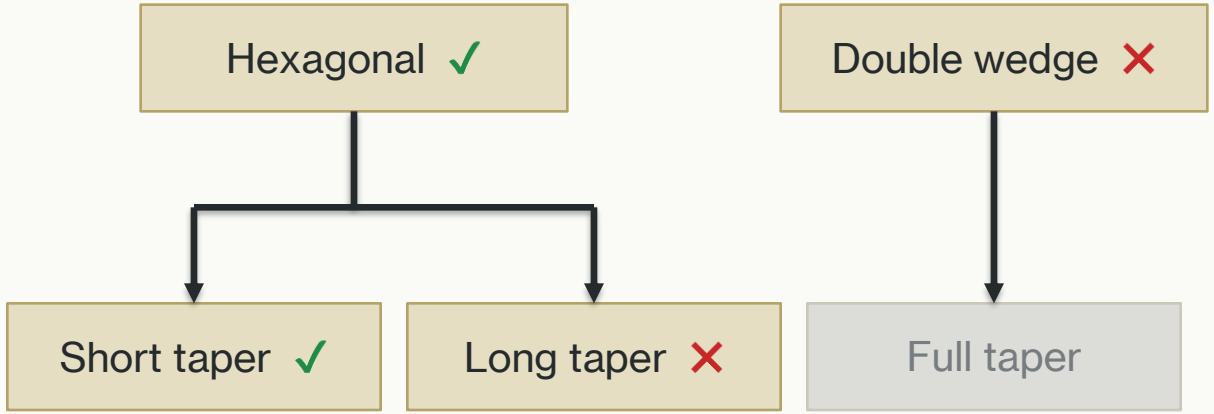
- Double wedge was aerodynamically preferred
- The alternative geometry carried only a small predicted apogee penalty
- Manufacturing repeatability became the primary discriminator

DECISION

- **Hexagonal**: repeatable to manufacture, higher predicted flutter margin; modest aerodynamic penalty accepted
- **Short taper**: easier machining and higher flutter margin
- Long taper rejected for machining difficulty; full taper dropped with the double wedge

VERIFICATION

- Simulation predictions: higher flutter margin, small apogee penalty



Detailed geometry optimization was performed later by the GTXR simulations team.

Design for reliability & feasibility

PROBLEM

- Select the core for the short-taper fin

CRITERIA

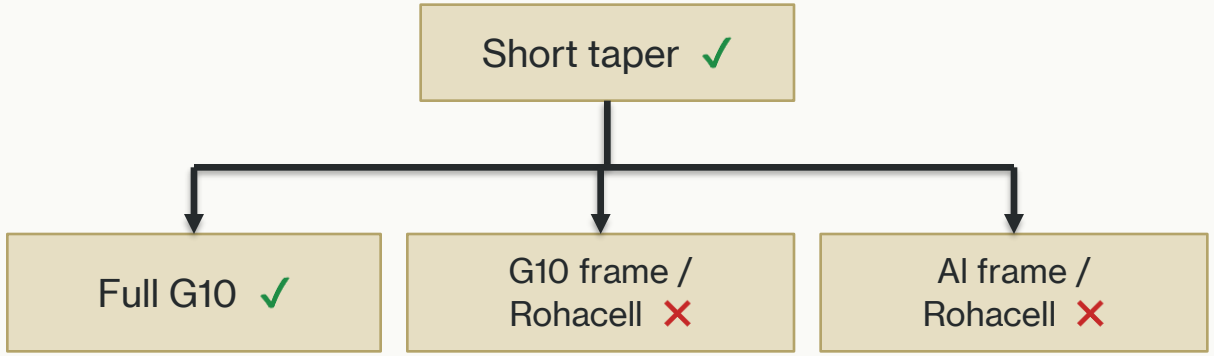
- Manufacturing confidence and repeatability across eight fins
- Mass
- Compatibility with the carbon-fiber skins

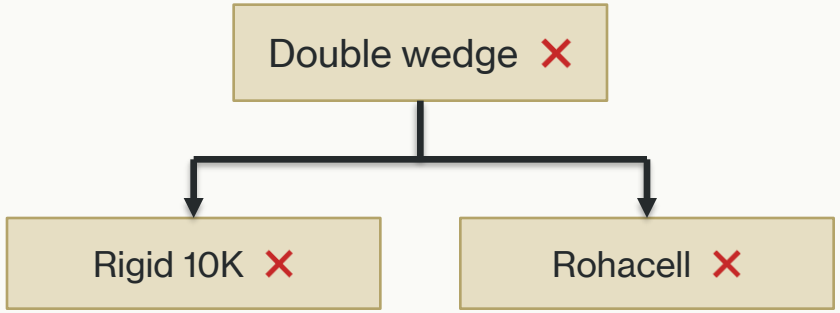
DECISION

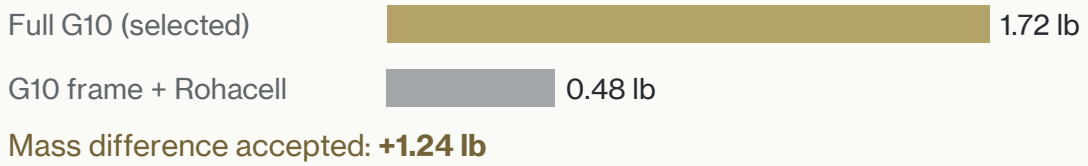
- ✕ G10 frame + Rohacell: 0.48 lb, lightest, but thermoforming was not yet mature or repeatable
- ✕ Al frame + Rohacell: CTE mismatch with carbon fiber; prior interface delamination
- ✕ Double-wedge cores: warping and immature processes
- ✓ **Full G10: 1.72 lb**, highest manufacturing confidence and repeatability

VERIFICATION

- +1.24 lb accepted for flutter margin, consistency and repeatability
- Follow-on: thermoformed Rohacell cores researched as a lighter alternative (AIAA 2026)







Designing a precision fin alignment jig

PROBLEM

- Four fins bonded at precise 90° spacing and 1.6° cant
- Alignment had to hold through bonding and cure
- Previous method: labor-intensive and hard to repeat

CRITERIA

- Clocking 90° ± 1°, from vehicle sensitivity analysis
- Cant 1.6° ± 0.1°
- Adapts to multiple rocket diameters and fin geometries
- Less setup and attachment time

DECISION

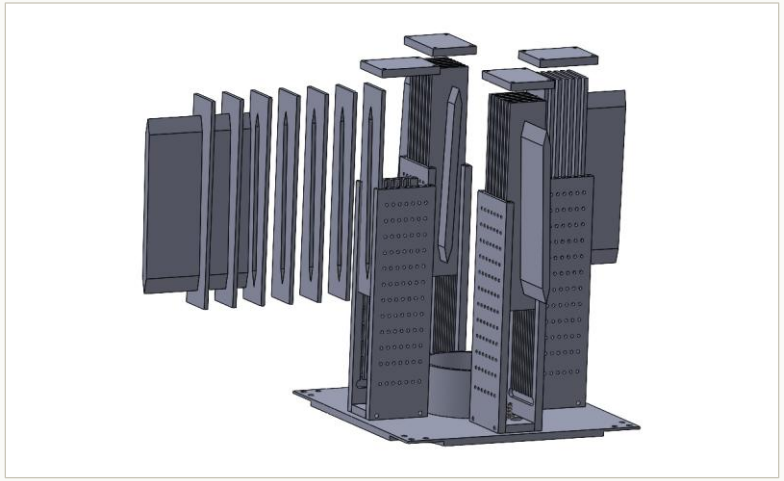
- Modular **four-tower fixture** designed around the fin-can geometry in SolidWorks
- All four fins positioned and bonded in one setup
- Led fixture design and the operating procedure; aided CAM programming and machining

VERIFICATION

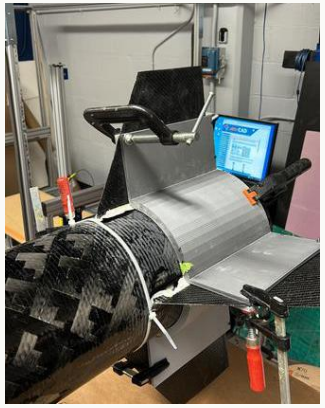
- ~1.5 hr labor saved; 24 hr cure-time reduction

NEXT

- Limitations identified and improvements designed; remachining planned for the next build cycle



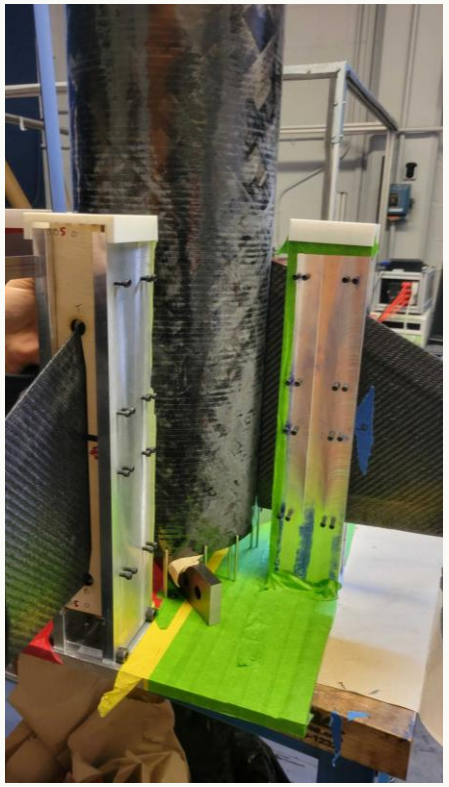
SolidWorks model: modular towers and base



Previous setup



Bonded fin with 1.6° cant



Fixture holding fins through bonding

Metrological 3D scan before jig

PROBLEM

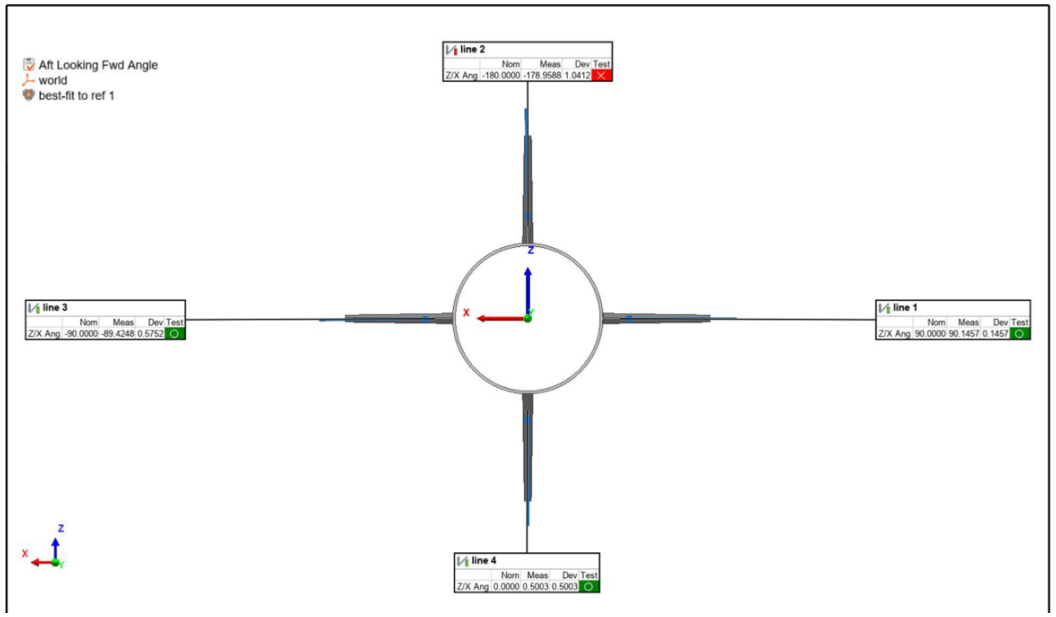
- Mini fin-can practice article, bonded with the previous method before the fixture existed
- 3D scan vs. nominal: 1 of 4 fins outside the ±1° clocking tolerance, at **1.041°**

CRITERIA

- Clocking 90° ± 1°, from vehicle sensitivity analysis
- Visual inspection alone could not verify alignment

DECISION

- Built the four-tower fixture; iterated the process and its usage instructions
- Standardized the layup and attachment sequence
- Improved layup application training
- Formal **3D-scan QC checkpoint** before flight hardware



Control View

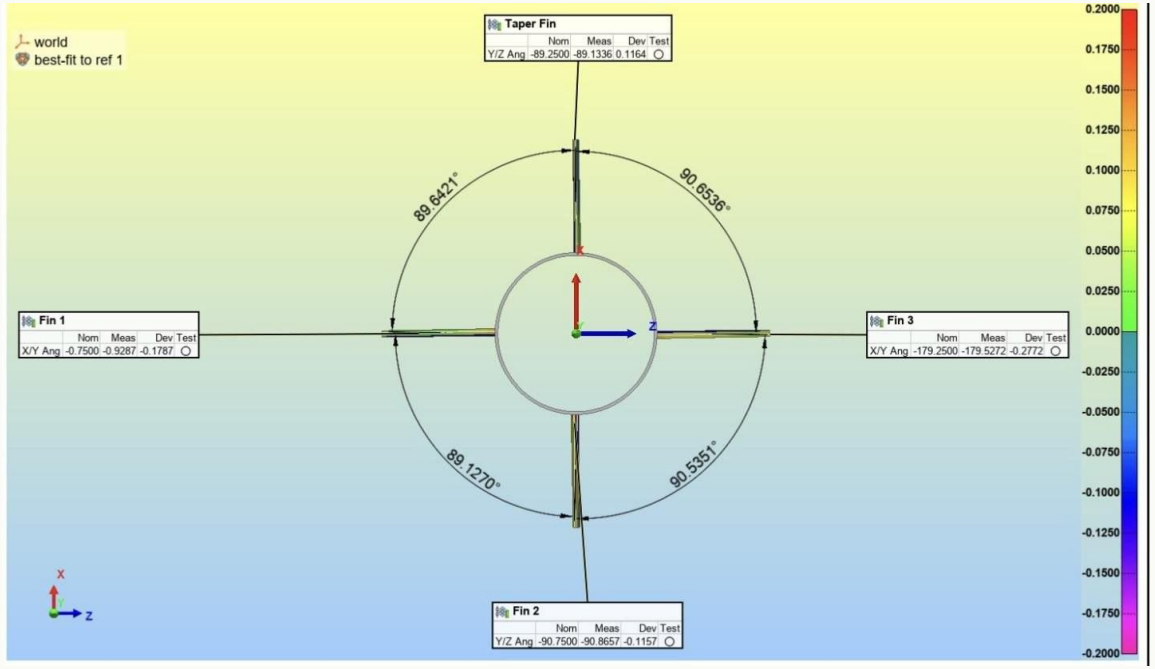
Control View Name	Aft Looking Fwd Angle
Units	Inches
Coordinate Systems	world
Data Alignments	best-fit to ref 1
All Statistics	Total: 4, Measured: 4 (100.0000%), Pass: 3 (75.0000%), Fail: 1 (25.0000%), Warning: 0 (0.0000%)

Char No.	Object Name	Control	Nom	Meas	Tol	Dev	Test	Out Tol
	line 1	Z/X Angle	90.0000	90.1457	±1.0000	0.1457	Pass	
	line 2	Z/X Angle	-180.0000	-178.9588	±1.0000	1.0412	Fail	0.0412
	line 3	Z/X Angle	-90.0000	-89.4248	±1.0000	0.5752	Pass	
	line 4	Z/X Angle	0.0000	0.5003	±1.0000	0.5003	Pass	

Practice article (mini fin-can) before the fixture: line 2 deviates 1.041° against a ±1° tolerance

Metrological 3D scan after jig

REQUIREMENT 1 · CLOCKING 90° ± 1°

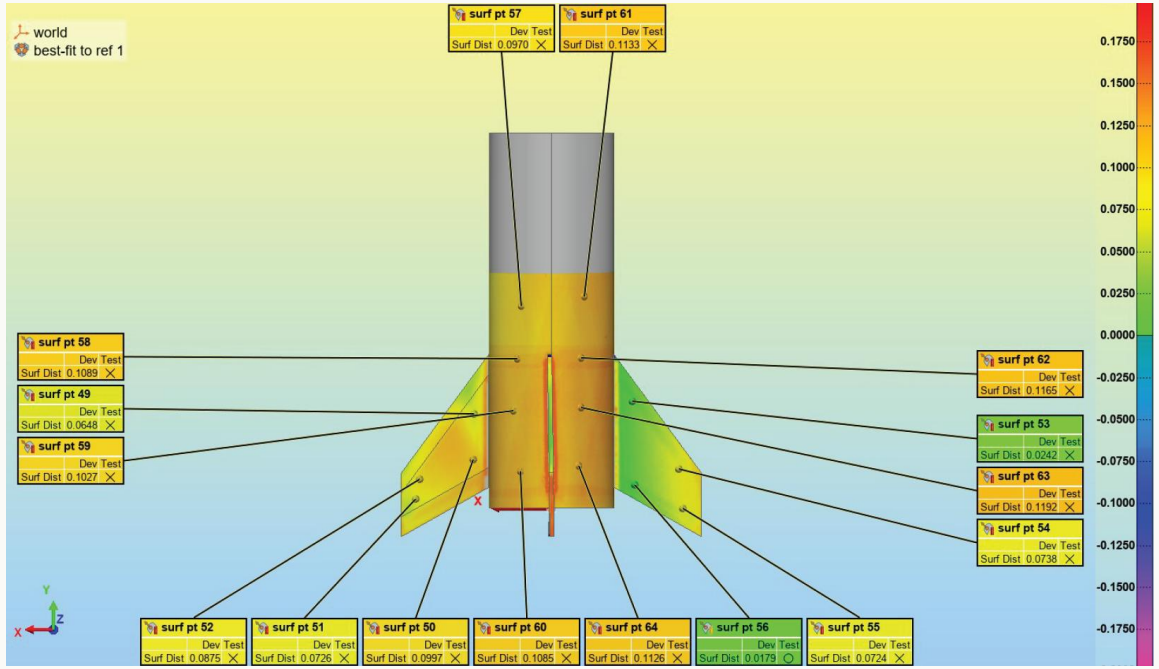


Practice article re-scan with the fixture (Fin 3 worst case: 0.277°)

RESULT

- Practice article: worst case 1.041° → 0.277° (~73% lower)
- Flight fins: clocking error **< 0.1°**

REQUIREMENT 2 · CANT 1.6° ± 0.1°



Practice article with the fixture: surface deviation vs. CAD (Hexagon)

RESULT

- Flight fins: all four within **0.1°** of the 1.6° cant target
- Fixture also saved ~1.5 hr of labor and 24 hr of cure time

Flight fins are not pictured: scans shown are of the practice article, made before the flight fins. The flight fins were scanned separately; results are from those scans.

Flight results

VERIFICATION

- Fin design cycle carried from trade study through build and inspection to flight hardware
- **Booster recovered** with fins intact in both vehicle cycles
- Sustainer had unrelated, non-structural anomalies, so its fins were not recovered



Recovered Mach N' Roll booster, fins intact



Mach N' Roll (2025)



Live and Let Fly (2024)

Plate-twist rig: validating shear modulus

I designed the plate-twist test apparatus · Co-author & presenter, AIAA 2024-85690 (7 authors)

PROBLEM

- Fin flutter predictions need the in-plane shear modulus of carbon-fiber/Rohacell sandwich panels
- Simulation-only values are hard to trust: process, bond quality, defects

CRITERIA

- Rigid plates sized to the 5 × 5 in panel; load points equidistant from center
- No local puncture of the panel
- Results within 5% of an ANSYS ACP model built by others

DECISION

- 1 in aluminum plates with ball-bearing contacts instead of puncture-prone cones
- Lower fixture rotated 90° to load each in-plane axis on an Instron 5982

VERIFICATION

- Three panels: 3.396 / 3.386 / 3.474 GPa vs. 3.281 GPa model
- **Mean within 5% of FEA** (≈ 4.2%)
- The validated value informed GTXR's fin optimizer

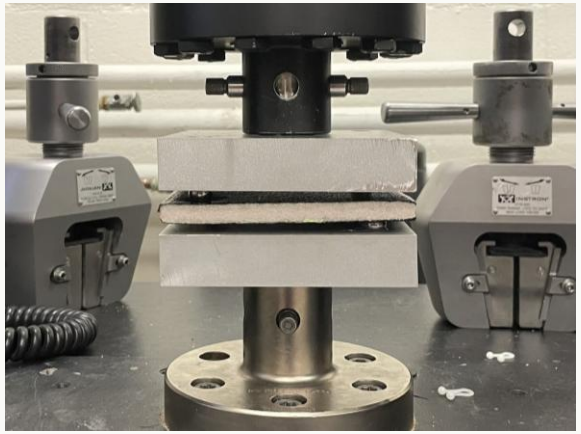
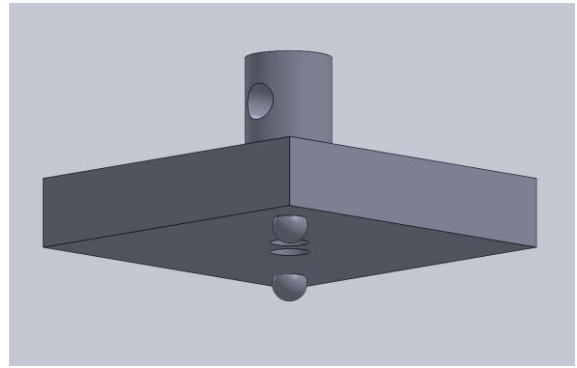
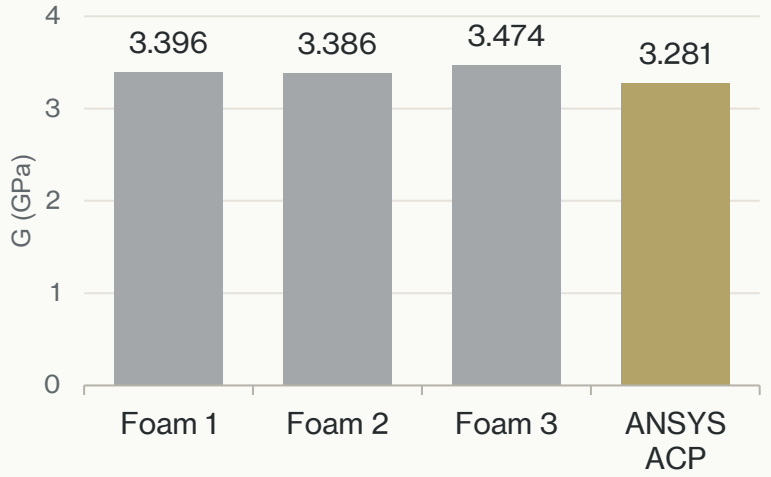


Plate-twist test on the Instron 5982



Rig CAD: ball-bearing contacts



≈ 4.2%

mean of three panels above the model

Mock launch rail improvements

PROBLEM

- The V1 rail flexed, was not straight, and did not hold the rocket steady

CRITERIA

- Stiff in bending and stable against buckling
- Straight and aligned while raising and holding the vehicle

DECISION

- With the structures lead, recommended an **I-beam rail** and a **larger base**
- Aided the design and load-path analysis
- Stay cables for alignment and raising

VERIFICATION

- Bending and buckling checks guided the design
- V2 in use for all-up practice



V1: rail flexed and drifted off straight



V2 during all-up practice

Multi-gas manifold design

Undergraduate Researcher · Jan 2026–present

PROBLEM

- Reacting jets-in-crossflow experiments need H_2, He, GOx, N_2 and natural gas delivered to predetermined inlets on the main flow rig

CRITERIA

- Regulated delivery of each gas to its assigned inlet
- Hydrogen supply combined from four bottles

DECISION

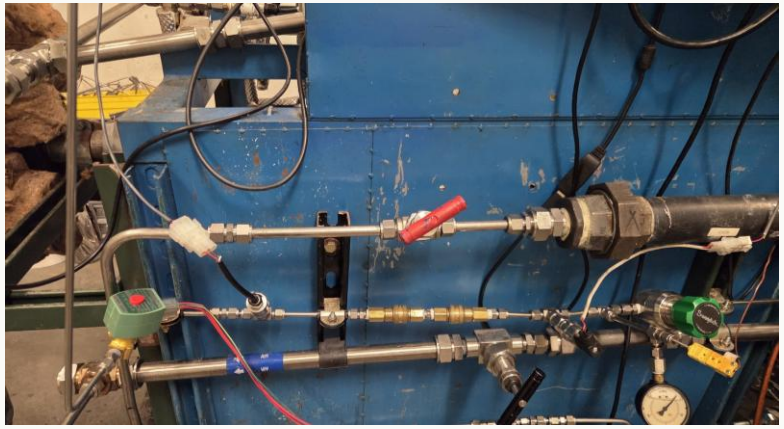
- **Designed and assembled** regulated multi-gas manifold hardware blocks
- Installed the distribution systems, including the piping and manifold tubing that combines four H_2 bottles

OUTCOME

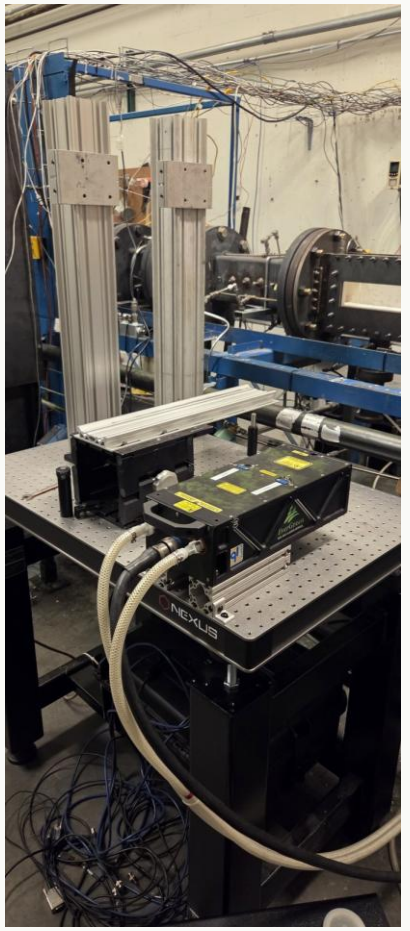
- Systems feed the main flow rig for reacting jets-in-crossflow experiments
- Helped construct the laser PIV setup; post-processed PIV data in MATLAB
- Now building hardware for a follow-on experiment



Tubing combining four H_2 bottles



Gas piping I installed



Laser PIV setup

Flight-critical design and build in 14 hours

Second-stage ignition passthrough, designed and built with the GTXR fin team

PROBLEM

- A flight-unit passthrough failed
- Second-stage ignition needed an electrical path through a pressure boundary

CRITERIA

- Seal at the NPT pressure-boundary interface
- Carry the electrical connection for second-stage ignition
- Pass a hydrostatic proof test at 2× MEOP before flight

DECISION

- Designed and fabricated a replacement flight-critical NPT passthrough with the fin team in **14 hours**

VERIFICATION

- Hydrostatic proof test at 2× MEOP, run by the propulsion lead
- Hardware flew; **the second stage ignited**
- The design informed later passthroughs



Finished NPT passthroughs

14 hr

failure to finished flight hardware

2× MEOP

hydrostatic proof (run by the propulsion lead)

Flew

second stage ignited

Thermoformed foam cores

Co-developed the process and machined the tooling · Co-author & presenter, AIAA 2026-112793 (3 authors)

PROBLEM

- Find a lighter alternative to dense fin cores: thermoform ROHACELL 51 into double-wedge cores

CRITERIA

- Geometric accuracy vs. CAD and repeatability across cores, measured by 3D scan

DECISION

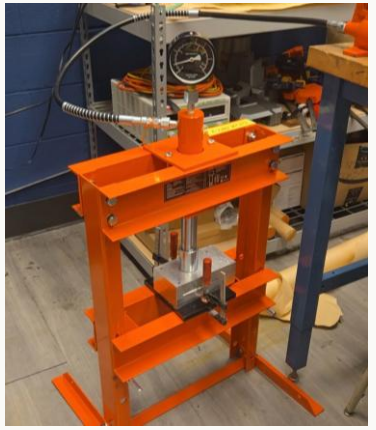
- Aluminum mold; blank heated to ~195 °C
- 4 t press load at the cavity centroid, not the geometric center, for planar contact
- Verdict: **promising, not yet flight-ready**; fixes proposed (mold borders / cutting inset, oversized blank)

VERIFICATION

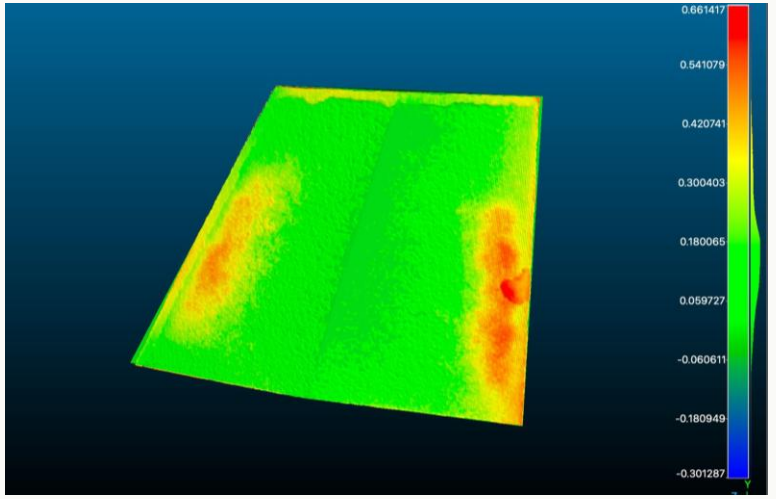
- 5 cores: 0.28 mm average global geometric error; 0.17 mm average surface std. dev.
- Error sources: mold thermal growth ≈ 0.59 mm (same order as the error), blank alignment, moisture, spring-back



Aluminum molds



Centroid-loaded press



Scan vs. CAD deviation, one formed core

0.28 mm
average global error, 5 cores

0.17 mm
average surface standard deviation

≈ 0.59 mm
mold thermal growth at forming temperature

Fixtures and gauges cut production times

Engineering Intern · Aug–Dec 2025 · Newburyport, MA

PROBLEM

- Marking load/unload took 6 s per part; inspection took 40 s

CRITERIA

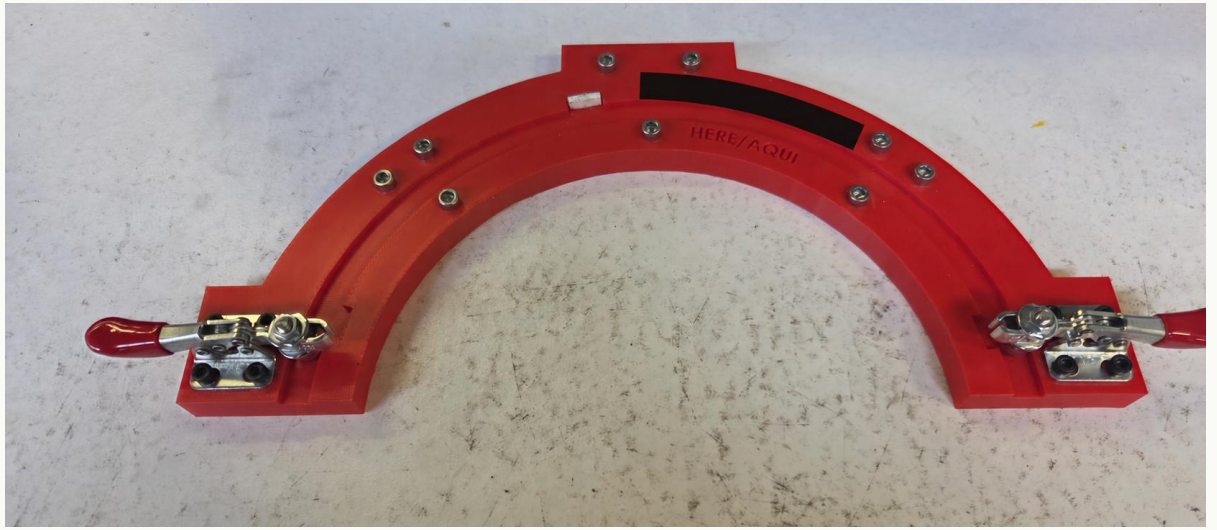
- Quick-change fixturing, designed with GD&T and DFM
- Fast, unambiguous inspection

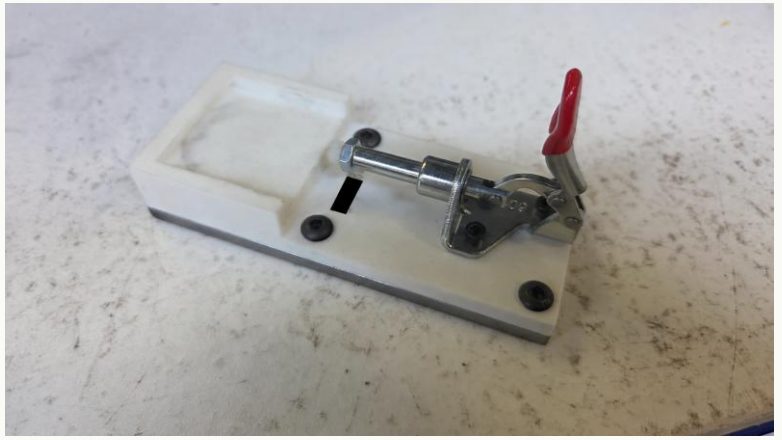
DECISION

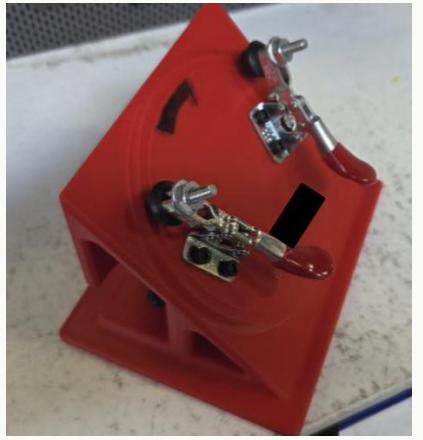
- 15 quick-change marking fixtures
- Go/No-Go gauges for inspection
- One-piece-flow changes

VERIFICATION

- Load/unload **6 s → 1 s** (83%); inspection **40 s → 2 s** (95%)
- Time between completed parts −55%







Marking fixtures. Identifiers removed. Go/No-Go gauges not pictured.

Qualifying an in-house cleaning process

Engineering Intern · Aug–Dec 2025 · Newburyport, MA

PROBLEM

- Precision cleaning of a Rolls-Royce component was outsourced, adding lead time, cleaning and shipping cost

CRITERIA

- Acceptable to Pursuit Quality and to Rolls-Royce
- Lower lead time and cost than the outside cleaner

DECISION

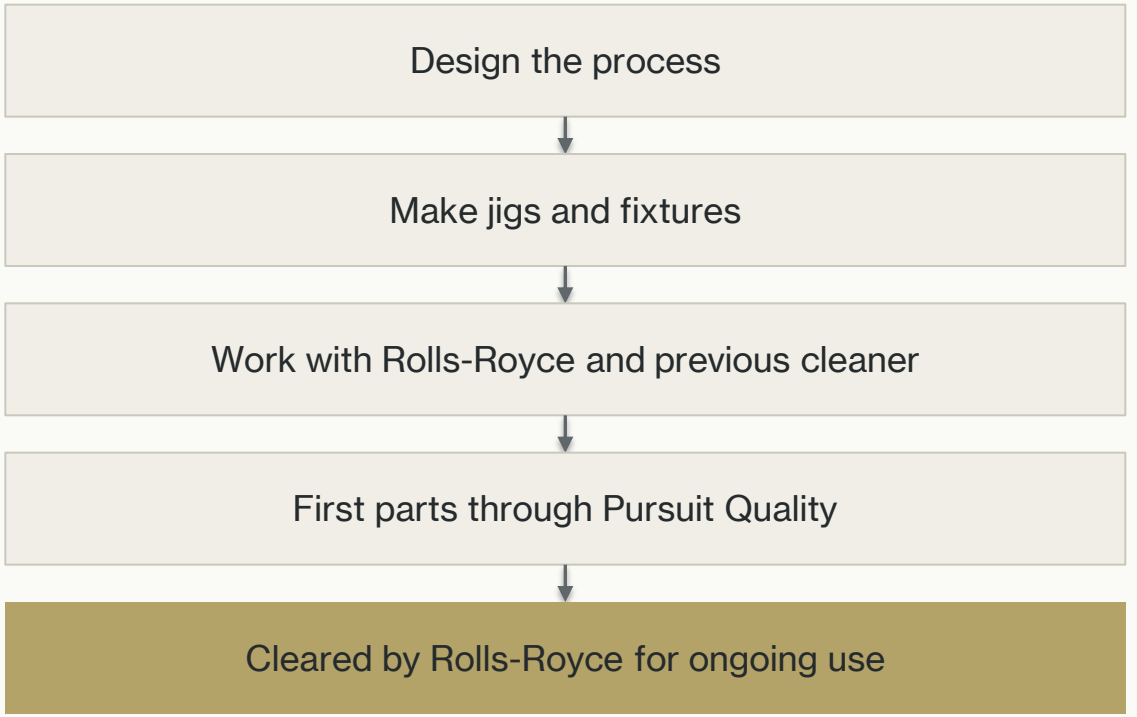
- Designed a new in-house cleaning process
- Made the jigs and fixtures for it
- Communicated with Rolls-Royce and the previous cleaner

VERIFICATION

- Took the first parts through Pursuit Quality
- **Cleared by Rolls-Royce** for ongoing use in Pursuit's shop

$1,000
saved per part

3 days
less lead time



Nathaniel Murcin

My work centers on mechanical and fluid systems design.

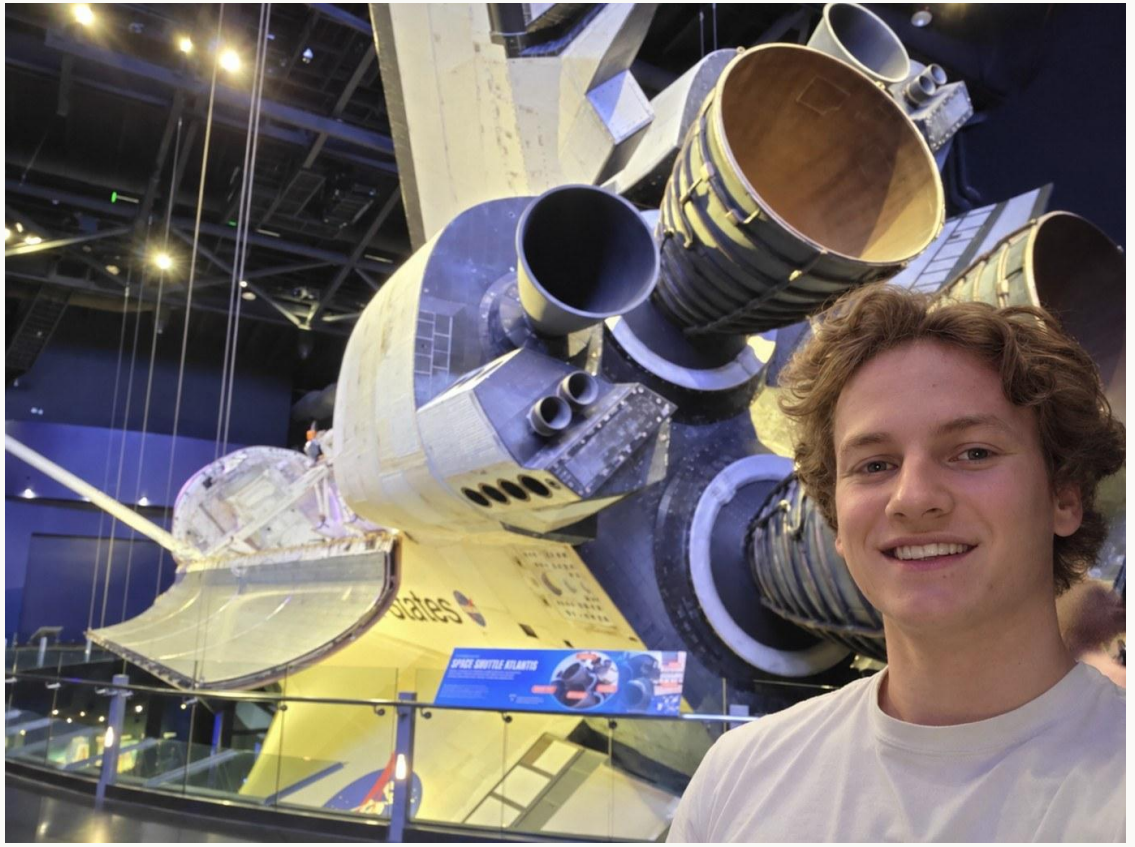
Aerospace Engineering · Georgia Institute of Technology

B.S. expected May 2027 · M.S. expected May 2028
Minor in Astrophysics · GPA 4.00/4.00

nmurcin3@gatech.edu

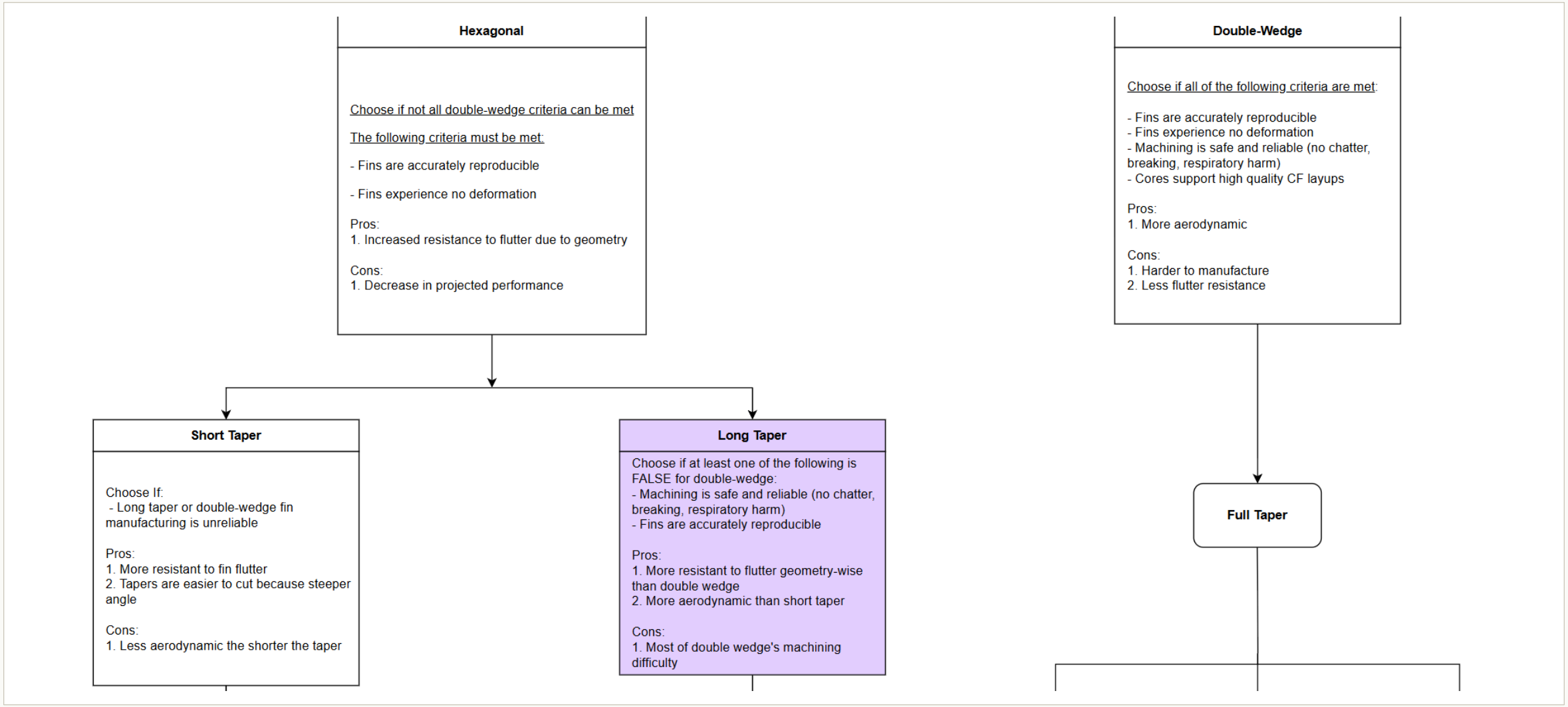
linkedin.com/in/nathaniel-murcin

nmurcin.github.io



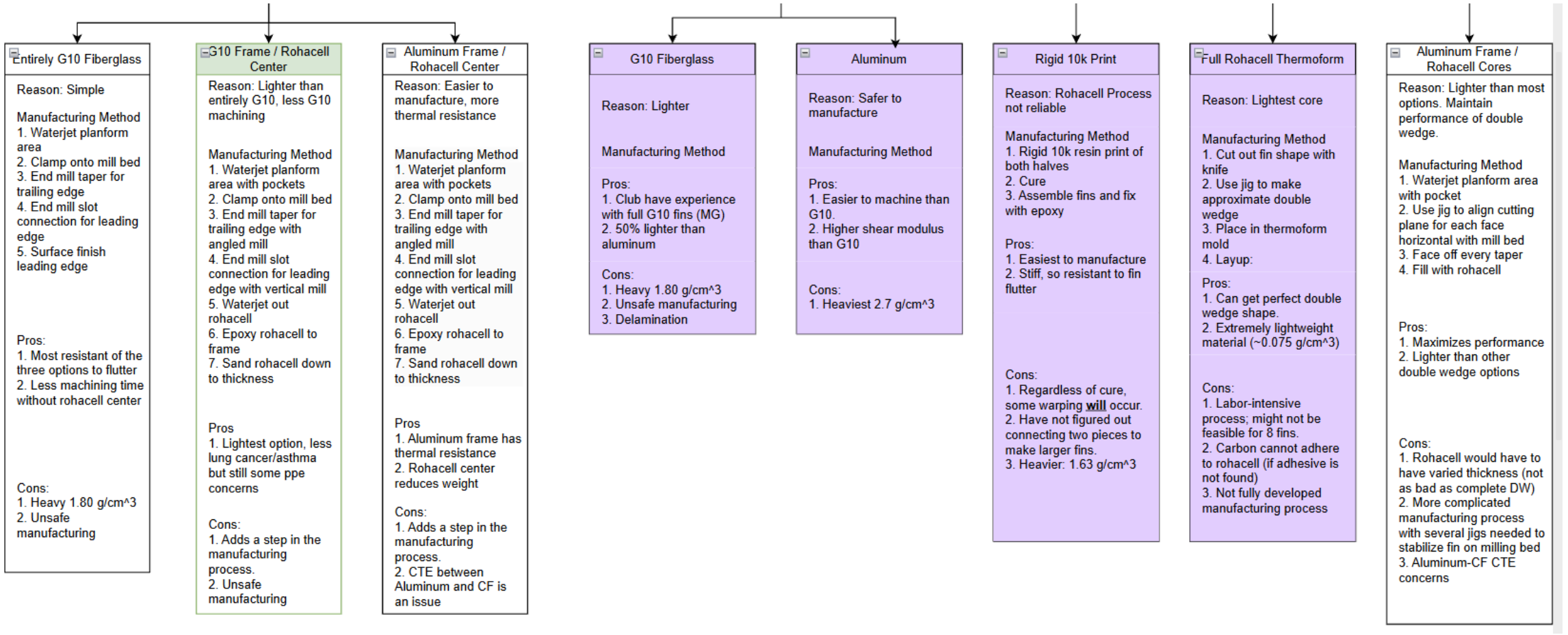
Space Shuttle Atlantis, Kennedy Space Center

Decision matrix: shape and taper



Purple boxes were eliminated. Zoom in for detail.

Decision matrix: core options



Purple boxes were eliminated. Zoom in for detail.